MATH 111 Calculus 1, NUU, FALL 2026

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The derivative at a point (Section 2.1, Lecture 7): secant and tangent slopes, the derivative from its definition, tangent lines, average and instantaneous velocity.

Key facts to have at hand (Lecture 7)

- **Secant slope = average rate of change** of f between a and $a+h$: $\dfrac{\Delta y}{\Delta x} = \dfrac{f(a+h)-f(a)}{h}$, the *difference quotient*. It depends on h.
- **The derivative of f at a** is the limit of the secant slopes, if this limit exists:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}.$$

  The two spellings are one number ($x = a+h$). Other names: $\frac{dy}{dx}\big|_{x=a}$ (Leibniz), $\frac{df}{dx}(a)$. (Lecture 7 writes x_0 for a, and Δx or h for the step.)
- **The tangent line** to $y = f(x)$ at $\big(a, f(a)\big)$ is the line through that point with slope $f'(a)$: $y - f(a) = f'(a)(x-a)$. It need not meet the curve only once.
- **The method** (Lecture 7's four steps). (1) Write $f(a+h)$ and open the brackets; (2) subtract $f(a)$ — for fractions, over a common denominator; (3) divide by h — allowed, because the limit never looks at $h = 0$ — and, for roots, multiply by the conjugate, until the h in the denominator cancels; (4) let $h \to 0$. Putting $h = 0$ into the quotient before the h has cancelled gives $0/0$ — an instruction to rewrite, not an answer.
- **Velocity.** If $s(t)$ is the position at time t, then $\dfrac{s(a+h)-s(a)}{h}$ is the *average* velocity over the time interval from a to $a+h$ (a secant slope), and $v(a) = s'(a)$ is the *instantaneous* velocity at time a (a tangent slope). The sign is the direction; the speed is $|v(a)|$.
- **Rates of change and units.** $f'(a)$ is measured in *units of f per unit of x*. Read aloud, $f'(a) = k$ says: *near a, one more unit of x changes f by about k units* — so $f(a+\Delta x) \approx f(a) + k\,\Delta x$ for small Δx.

Warm-up · True or false? (justify with one sentence or one counterexample)

1. A tangent line to a curve meets the curve at exactly one point.
2. $\dfrac{f(3)-f(1)}{2}$ is the slope of the secant line through $\big(1, f(1)\big)$ and $\big(3, f(3)\big)$.
3. $\lim\limits_{h\to 0} \dfrac{f(5+h)-f(5)}{h}$ and $\lim\limits_{x\to 5} \dfrac{f(x)-f(5)}{x-5}$ can be two different numbers.
4. If $f'(2) = 0$, then the tangent line to $y = f(x)$ at $x = 2$ is horizontal.
5. If $s(t)$ is the position (in metres) at time t (in seconds), then $\dfrac{s(5)-s(2)}{3}$ is the velocity at $t = 5$.
6. If $C(x)$ is the cost in dollars of producing x kilograms, then $C'(100)$ is measured in dollars.
7. If $f(3) = 0$, then $f'(3) = 0$.
8. If $f(x) = 7x - 4$, then $f'(a) = 7$ for every number a.

Problem 1 *The derivative from its definition, and the tangent line.*

Find $f'(a)$ from the definition, writing out the four steps; then find an equation of the tangent line to $y = f(x)$ at the point $\big(a, f(a)\big)$.

(a) $f(x) = 3x^2 - 5x, \;\; a = 2$

(b) $f(x) = x^3 - 2x, \;\; a = 1$

(c) $f(x) = \dfrac{2}{x+1}, \;\; a = 1$

(d) $f(x) = \sqrt{2x+1}, \;\; a = 12$

(e) $f(x) = \dfrac{x}{x-2}, \;\; a = 3$

Problem 2 *Average and instantaneous velocity.*

A ball is thrown upwards from the edge of a balcony; it rises, then falls past the balcony to the ground. Its height above the ground t seconds after the throw is

$$s(t) = 20 + 15t - 5t^2 \text{ metres} \qquad \text{(until it lands)}.$$

(a) Find the average velocity of the ball over the time intervals $[1, 2]$, $[1, 1.5]$, $[1, 1.1]$ and $[1, 1.01]$. What number do these averages approach?

(b) Find the instantaneous velocity at $t = 1$ from the definition. Is the ball going up or down?

(c) Find the velocity at an arbitrary time $t = a$. When is the ball at its highest point (where it is momentarily at rest), and how high is it then?

(d) When does the ball hit the ground, and how fast is it going at that instant?

(e) Find the average velocity over the whole flight, from the throw to the landing, and compare it with the velocity at impact.

Problem 3 *Tangent lines with a condition.*

Let $f(x) = x^2 - 4x + 5$.

(a) Show from the definition that $f'(a) = 2a - 4$ for every number a. At which point of the graph is the tangent line horizontal?

(b) Using the formula of (a), find the point at which the tangent line is parallel to the line $y = 2x + 1$, and write the equation of that tangent.

(c) Find every tangent line to the graph that passes through the point $(0, 1)$. (The point is not on the graph.)

Problem 4 *Which function, which point?*

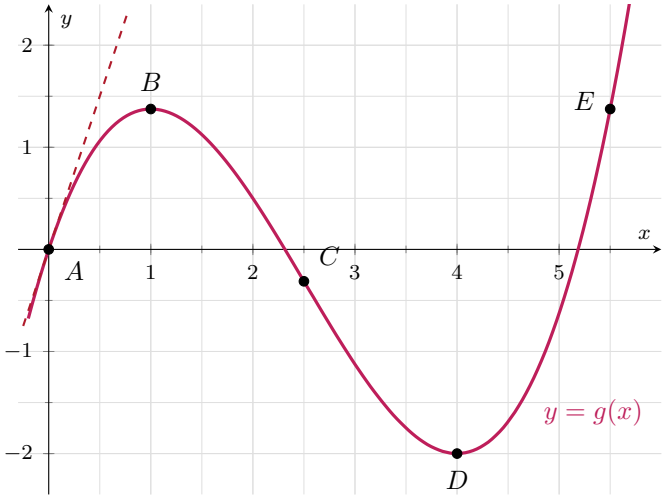
Each limit is the derivative $f'(a)$ of some function f at some number a. Say which f and which a, and find the value of the limit.

(a) $\displaystyle\lim_{h\to 0} \frac{(3+h)^2 - 9}{h}$ (b) $\displaystyle\lim_{x\to 4} \frac{\sqrt{x} - 2}{x - 4}$ (c) $\displaystyle\lim_{h\to 0} \frac{\frac{1}{2+h} - \frac{1}{2}}{h}$ (d) $\displaystyle\lim_{x\to 1} \frac{x^4 - 1}{x - 1}$

(e) Find a second pair (f, a) that answers (a).

Problem 5 *Reading slopes off a graph.*

The graph of a function g is shown, with five marked points A–E; the dashed line is the tangent to the graph at A.



(a) At which of the points A–E is the derivative positive? negative? zero?

(b) Use the dashed tangent line to find $g'(0)$.

(c) Arrange in increasing order: 0, $g'(0)$, $g'(2.5)$, $g'(5.5)$, and the slope of the secant line through A and E.

(d) (A different function.) The tangent line to $y = k(x)$ at the point $(2, 5)$ passes through $(6, -3)$. Find $k(2)$ and $k'(2)$.

Challenge *(for those who finish early, or at home)*

1. Find, from the definition, the slope of the tangent to $y = x^3$ at $(1, 1)$, and the equation of the tangent. Show that this tangent meets the curve at a second point, and find it.
2. Take $x_0 > 0$ and the point $P = \left(x_0, \frac{9}{x_0}\right)$ on the hyperbola $y = \frac{9}{x}$. Show that the tangent at P cuts off from the first quadrant a triangle of area 18, wherever P is, and that P is the midpoint of the part of the tangent that lies between the axes. (Lecture 7 did this for $y = \frac{1}{x}$.)